

PROJECT PROPOSAL DATA STRUCTURE

Project title: Sudoku Solver

Name: Mishal Fahim (k24-0041)

Course Instructor: Shafique Rehman

Sudoku Solver Using Backtracking (C++)

1. Introduction

Sudoku is a logic-based puzzle where a 9x9 grid must be filled with digits 1–9 following specific rules. This project implements an automated solver using recursion and backtracking.

2. Objectives

- Implement a Sudoku solver using C++.
- Apply recursion and backtracking.
- Validate Sudoku constraints efficiently.

3. Problem Description

Each row, column, and 3x3 subgrid must contain digits 1–9 without repetition. The program fills empty cells (represented by 0).

4. Methodology

The solver uses recursion to try possible numbers in empty cells. If a placement leads to a dead end, the algorithm backtracks and tries a different number.

5. Algorithm (Pseudo Code)

```
solveSudoku(grid): if no empty cell: return true
for num = 1 to 9: if safe: place number if
solveSudoku: return true
remove number
return false
```

6. Data Structures

A 2D array is used to represent the Sudoku grid. Functions include `isSafe()` and `solveSudoku()`.

7. Example Input

```
5 3 0 0 7 0 0 0 0
6 0 0 1 9 5 0 0 0
...
```

8. Expected Output

5 3 4 6 7 8 9 1 2
6 7 2 1 9 5 3 4 8
...

9. Time Complexity

Worst-case exponential complexity due to backtracking.

10. Future Enhancements

GUI interface, puzzle generator, difficulty levels.

11. Conclusion

This project demonstrates recursion and efficient problem-solving techniques in C++.